

Generative Engine Optimization at Scale: Measuring Brand Visibility Across AI Search Engines

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Abstract

The traditional blue-link era has evolved significantly over the past few years. Rather than typing queries into traditional search engines like Google and Bing, people have shifted to asking on AI search engines like ChatGPT, Claude, and Gemini. As a result, brands that once concentrated on search engine optimization (SEO) must now also work to optimize and position themselves effectively within AI search engines. A set of new and evolving terms describes this shift: Generative Engine Optimization (GEO), Answer Engine Optimization (AEO), and AI Search Visibility. All of them focus on how brands are represented, cited, and recommended by AI search engines. These AI engines rely on large language models (LLMs) to gather and process the vast range of information available across the internet. We treat AEO and AI Visibility as part of the broader GEO scope.

In this paper, we explain how to measure brand visibility and analyze responses across AI search engines. We look at what these engines appear to value when they cite brands, what types of sources they rely on, and what kind of content LLMs are more likely to process and surface. The top brands that are already authoritative tend to get cited naturally. The harder and more important problem is for everyone else, including SMEs, D2C brands, creators, early-stage startups, and brands without much of an online footprint.

We demonstrate our findings by analyzing 100K+ prompt responses across 100+ brands tracked on Ranqo between March and May 2026. The very first visibility runs form a clear three-tier brand-stature ladder: global household names such as Stripe and Nike appear in 73% of relevant AI answers on their first tracking run; established mid-market and regional brands such as Olipop and Klaviyo appear in 44%; niche and small brands in just 11% — each tier about 30 percentage points below the last. When the AI search engines cite any sources, about 78% of those citations go to corporate websites, including the brand’s own pages and third-party pages from companies in the same space. Among non-corporate sources, YouTube is cited most often — ahead of Reddit, editorial media, and Wikipedia. At the page level, the highest-leverage citation surface is the ranked “best-of” listicle, the most-cited content format at about 21% of all citations: a single “best X tools” page can surface a brand across many AI answers at once. Sentiment is the unstable part: whether the AI frames a brand positively or negatively flips about 6.7 times more often than whether the brand is mentioned at all.

Together, these findings provide a first large-scale baseline for measuring GEO. They show that AI brand visibility can be measured, differs by platform, and varies strongly by brand maturity. We end the paper by proposing seven v1.1 protocols to test whether specific recommendations can causally improve AI visibility.

We wrote this paper for a broad audience: people who are just getting started with GEO, AEO, and AI visibility, as well as practitioners, founders, marketers, and researchers who already work in this space and want to understand the field in more depth. Our goal is to make the paper useful for anyone trying to measure, compare, and improve brand visibility across AI search engines.

Keywords: Generative Engine Optimization, GEO, Answer Engine Optimization, AEO, AI Search Visibility, AI Visibility, AI-optimized content, large language models, AI search engines, citation analysis, brand visibility, share of voice, ChatGPT, Claude, Perplexity, Gemini, Grok, E-E-A-T, multi-platform measurement, Ranqo, Ranqo AI.

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1 Introduction

More and more, people get their answers straight from an AI assistant instead of scrolling through ten blue links. They ask ChatGPT, Claude, Perplexity, Gemini, or Grok what to buy, what tool to use, or who to trust, and they act on what comes back. For a brand, this quietly changes the question that matters. The old question was whether you rank for a keyword. The new question is whether an AI model, when someone asks about your category, actually names you.

This is what GEO is about. Generative Engine Optimization (GEO) is the practice of measuring and improving how AI search engines represent, cite, and recommend a brand (Aggarwal et al., 2024). Two related terms get used alongside it: Answer Engine Optimization (AEO) and AI Search Visibility. We treat both as part of the broader GEO scope in this paper, because they describe the same job from slightly different angles. GEO is also not a replacement for SEO. It is built on top of good SEO, since the content an AI engine ingests is, for the most part, the content you already publish for search.

There is one more thing worth saying up front. The brands that are already big and authoritative tend to get cited no matter what. The hard, unsolved problem belongs to everyone else: the SMEs, the D2C brands, the creators, the early-stage startups, and any brand without much of an online footprint. That is the audience we care about most, and the gap this paper sets out to measure.

A growing body of academic work already tells us that the AI-search playbook is not the old SEO playbook. Generative engines pull from a broader and very different set of sources than classical search, and they lean on their own internal knowledge to different degrees (Kirsten et al., 2025). When they do cite, the citations concentrate on a small set of outlets, and different engines pick different outlets (Yang, 2025). LLMs also reward sources that are already widely cited, a Matthew effect that makes the rich richer (Algaba et al., 2025). And the first careful benchmark of “conversational SEO” tactics found that most of them do not help, and several actively hurt (Puerto et al., 2025). We build on these findings with production data.

What this paper contributes.

1. **A way to measure AI brand visibility in production.** We describe Ranqo (also Ranqo AI), a platform that issues controlled queries to five AI engines and measures, for each brand, whether the engines mention it, where they rank it, how they frame it, and its share of voice against competitors. It classifies the sources behind those answers, runs a six-dimension page audit, and re-measures on a recurring cadence.
2. **A category reference study.** We summarize Ranqo’s published CRM study (Ranqo, 2026b): 50 unbranded prompts sent to all five engines ten times each (2,500 responses) across ten CRM brands.

3. **Findings from production data.** A day-1 recognition finding (§6.2); mention behavior that is mostly deterministic, with flat trajectories absent any intervention (§6.4); a source breakdown showing the brand’s own site is a small slice and most citations go to other companies (§6.5); sentiment that is far noisier than mention (§6.6); the brand-stature visibility ladder, which is the headline result (§6.1).

4. **A practical read, especially for smaller brands.** We turn the findings into plain guidance for the brands that do not get cited automatically, and we propose seven v1.1 protocols to test whether specific recommendations causally move AI visibility.

The rest of the paper is laid out as follows. §2 covers the related work briefly. §3 describes the Ranqo platform and §4 the measurement methodology. §5 gives the CRM reference study. §6 presents the findings and the v1.1 protocols. §7 discusses what they mean, §8 lists the limits, and §9 concludes.

2 Background and Related Work

2.1 SEO, AEO, GEO, and AI Visibility

A handful of terms now describe overlapping work, so we fix what we mean by each, following the practitioner usage in (Ranqo, 2026a,c). **SEO** is about blue-link ranking on classical search engines. **AEO** is about being the single source a featured snippet or direct-answer box pulls from. **GEO** is about being cited and recommended when a large language model synthesizes an answer across many sources. **AI Visibility** is the broader question of whether an entity is represented inside a model’s parameters and retrieval indexes at all. In this paper we use GEO as the umbrella term and keep AEO and AI Visibility under it, because for a brand the practical surface is the same: what the engines ingest, and what they say back.

2.2 What the research already shows

The academic picture is young but consistent on a few points, and we lean on it throughout.

The foundational benchmark is the Princeton GEO study (Aggarwal et al., 2024) (KDD 2024), which ran 10,000 queries and tested content changes for their effect on whether an answer cites a page. The biggest wins came from adding machine-extractable provenance: quotations, statistics, and citations, each worth roughly 25–40% more visibility. That is still the core of the modern GEO playbook.

The most useful corrective is C-SEO Bench (Puerto et al., 2025), the first systematic benchmark of “conversational SEO” tactics. Its headline is blunt: most of those tactics do not help, and several hurt, while plain source relevance keeps working. This shapes how we run our own experiments. We test changes to page-quality signals on the source side, not prompt-engineering tricks on the response.

On how engines actually cite, three recent studies line up with what we see. Kirsten et al. (2025) show generative engines pull from a broader and very different source set than classical search, and rely on their own internal knowledge to varying degrees. Yang (2025), across roughly 366,000 citations, find that citations concentrate on a small set of outlets and that different engines pick different ones, so cross-platform agreement is low. Algaba et al. (2025) show LLMs prefer sources that are already heavily cited, a Matthew effect. Together these explain a pattern we keep coming back to: the brands with the most existing web presence get cited the most, which is exactly the hill smaller brands have to climb.

3 The Ranqo Platform

3.1 Platform query layer

For each tracked brand, Ranqo generates a set of natural-language prompts spanning six categories (*discovery*, *problem_solution*, *comparison*, *use_case*, *expert*, and *brand_research*) and issues each prompt to all five engines via official APIs. Table 1 summarizes the per-engine configuration.

Model selection. For each engine we query its current search-enabled production model (Table 1) — the variant that exposes web retrieval through the normal API. Gemini, Perplexity, and Grok are grounded on every query; ChatGPT and Claude run web search on a weekly cooldown per brand to control cost, and between those passes they answer from their parameters alone — the search-on/search-off contrast we use in §6.1. Holding the search-enabled variant constant across engines keeps the setup comparable, so the cross-engine results in §6.1 reflect how the engines ingest and retrieve rather than how they bundle search; how far the patterns generalize to other model configurations is left to future work (§8). Per-brand location is passed as a native geo parameter where the API supports one (ChatGPT, Claude, Perplexity) and as a system-prompt suffix otherwise (Gemini, Grok).

3.2 The six prompt categories

The six categories are not interchangeable. A brand can look strong in one and be almost invisible in another, so we track all six separately. Using running shoes as the example throughout:

Discovery Broad, top-of-funnel questions where the buyer is still exploring and names no brand. “best running shoes,” “good running shoes for beginners.”

Problem/solution The buyer has a specific problem and wants something that fixes it. “I have ankle pain, what kind of running shoes should I wear,” “shoes for flat feet.”

Use case A specific activity or setting. “best shoes for trail running,” “shoes for standing all day at work.”

Comparison Weighing named options against each other.

“Nike Pegasus vs Brooks Ghost,” “Hoka or Asics for a marathon.” These usually name brands.

Expert Deeper, more technical questions from an informed buyer. “what heel-to-toe drop suits a stability shoe,” “does a carbon plate actually make me faster.”

Brand research Questions about one named brand. “is Nike a good running brand,” “Nike running shoes reviews.” These name the brand by definition.

The split matters for measurement. The first four (discovery, problem/solution, use case, and expert) are normally unbranded, so they test whether an engine surfaces the brand when nobody has named it. That is the hard, GEO-relevant case, and it is the one we lean on throughout the paper. Only comparison and brand research name a brand, which makes a mention much more likely and tells us far less about discovery. For that reason the starter set we generate for a new brand is deliberately unbranded.

Prompt themes. Categories describe the shape of a single prompt. *Themes* sit one level up. A theme is a topical cluster (a product line, a use-case area, a competitor set, a buying occasion), and the platform generates a batch of prompts under each theme while keeping the category mix balanced inside it. Themes let a brand track a whole topic at once (say, “trail running shoes” or “shoes for nurses”) instead of writing prompts one at a time, which is how tracking scales from a handful of prompts to broad topic coverage.

3.3 Citation extraction

For each engine response, Ranqo extracts:

- **Brand-mention signals:** a Boolean `brandMentioned`, an integer `brandPosition` (1st, 2nd, 3rd, ...), a sentiment label (positive / neutral / negative) accompanied by a continuous `sentimentScore` $\in [-1, 1]$ and an array of sentiment-bearing text spans.
- **Source citations:** every URL the engine grounded the answer on, normalized to canonical domain and classified into a type (`news_media`, `community`, `review`, `video`, `social`, `developer`, `academic`, `government`, `reference`, `corporate`).
- **Source-to-brand relationship:** each cited source is labeled as *you*, *competitor*, *non_competitor*, *editorial*, *forum*, *review*, *social*, *reference*, *institutional*, or *other*.
- **Competitor mentions:** an array of competitor names with their ordinal positions.

All signals are stored at the level of a single (prompt, platform, run) tuple.

3.4 Scoring methodology

Each (prompt, platform, run) tuple carries three brand signals: a Boolean mention, an ordinal position when

Table 1. Engine configuration: model (class, not version), web-search policy, and citation surface.

Engine	Model	Web-search policy	Citation surface
ChatGPT	GPT-5	Weekly cooldown per brand	annotations.url_citation
Gemini	Gemini-3	Always grounded	groundingChunks[].web
Perplexity	Sonar	Always grounded (search-native)	search_results array
Claude	Claude Sonnet	Weekly cooldown per brand	text-block citations[]
Grok	Grok-3	Always-on retrieval	citations[] + inline markdown

mentioned, and a sentiment label. The three behave very differently — mention is stable while sentiment is far noisier (§6.6) — so we report them as separate metrics rather than collapse them into one weighted score.

1. **Visibility (the headline metric).** Per engine, the mention rate: the share of a brand’s prompts on which it is mentioned, mentioned/total $\times 100$. The cross-engine figure is the platform-weighted average of the per-engine rates — ChatGPT 0.30, Gemini 0.20, Perplexity 0.20, Claude 0.20, Grok 0.10, renormalized over the platforms a brand actually runs on so plan-based subsets still sum to 1.0.
2. **Average position.** The mean ordinal rank on the prompts where the brand is mentioned (1 = first recommendation), weighted across engines the same way.
3. **Sentiment score.** (positive+0.5 neutral)/mentions $\times 100$ — all-positive 100, all-neutral 50, all-negative 0, so a negative mention costs twice a neutral one.
4. **Share of voice.** The brand’s mentions divided by the brand’s plus its competitors’, counting only competitors that recur or resolve to a real domain so one-off names an engine hallucinates do not inflate the denominator.

Metrics are computed per completed run; when a single query fails its previous value is carried forward and flagged, so a transient API error is not read as a visibility drop. Prompt categories drive the per-category breakdown (§6.3) but are not weighted into the aggregate. The 0–100 visibility score maps to letter grades $A+ \geq 90$, $A \geq 80$, $B+ \geq 70$, $B \geq 60$, $C+ \geq 50$, $C \geq 40$, $D \geq 30$, $F < 30$. Every coefficient lives in configuration rather than being hardcoded, so the scoring is transparent and straightforward to reweight.

3.5 Six-dimension page audit

In parallel with the query layer, Ranqo runs a six-dimension page audit against each tracked URL, scoring each dimension 0–100 and combining them into a weighted overall page score:

1. **Crawlability** — whether an AI crawler can reach and parse the page at all: robots, sitemap, canonical, HTTPS, JS rendering, and mobile-friendly checks.
2. **Content Quality** — whether the page is substantive and well-structured: word count, readability, paragraph

structure, heading hierarchy, internal linking, and freshness.

3. **Page Speed** — how quickly the page loads for users and crawlers: Core Web Vitals (LCP, FID, CLS).
4. **AI Readiness** — how easily a model can lift a direct answer from the page: entity clarity, structured answers, inline citations, evidence density, and provenance markers.
5. **Citation Potential** — whether the page offers something concrete worth quoting: quotability units (numbers, definitions, named claims), source attribution, and data availability.
6. **Authority & Trust** — whether the page signals credible, first-hand expertise: visible author byline, credentials, domain authority, topical depth, and E-E-A-T markers (Ranqo, 2026h).

Together, the three GEO-specific dimensions (AI Readiness, Citation Potential, and Authority & Trust) carry the largest share of the page score and are the ones most specific to GEO: they score whether a page states its facts clearly, backs them with quotable evidence, and makes its authorship visible. These line up with the content-provenance signals the Princeton benchmark found most effective (Aggarwal et al., 2024).

3.6 Closed-loop recommendation engine

For every brand, Ranqo’s recommendation engine (Claude Sonnet, temperature 0.5, with structured-output validation) ingests visibility trends, category performance, content gaps, competitor threats, sentiment distribution, and optional GA4 traffic context, and emits a ranked list of ImprovementRecommendation rows. Each row carries a category, an action type, a priority class, and two scores on a 1–3 scale (impact, effort).

Each recommendation also carries baselineScore, currentScore, and scoreDelta on the dimension it targets. When the recommendation is implemented and re-audited, the delta becomes observable. This is the closed-loop primitive the v1.1 RCT (Protocol P3, §6.7) is built on.

3.7 From diagnosis to AI-optimized content

Diagnosis is only half of a closed loop. Where the recommendation engine says *what to fix*, a content pipeline turns the same diagnostic into *what to publish*: it reads

the identical context — visibility gaps (prompts on which the brand never surfaces), competitor advantages, weak prompt categories, and page-audit content gaps — and emits ranked content briefs (a title, an angle, the exact target prompts addressed, and a priority), weighted toward the topics where the brand is currently invisible rather than generic expansion.

Across this workflow, role-specialized agents each handle a different task: a briefing agent ranks topics from the diagnostic, a drafting agent writes the article from the brief, a humanization agent rewrites it for a natural voice, and an optimization agent scores an existing page against the audit dimensions and proposes targeted edits; a deterministic stage then attaches an Article + FAQPage JSON-LD graph. Every agent is engineered to the page-quality signals from §3.5 that the Princeton benchmark found most effective (Aggarwal et al., 2024) — an entity-first opening, quotable statistics and named claims, an FAQ block, and structured headings — so the output is built to be cited by AI engines, not merely to read well.

This is the *act* half of that closed loop: diagnose where a brand is invisible, publish content built to fix it, then re-measure. We describe the mechanism here; whether the generated content causally lifts AI citations is the closed-loop question we leave to the v1.1 RCT (Protocol P3/P7, §6.7) rather than assert it.

3.8 Data scale and the unit of replication

The platform’s unit of replication is a (brand, prompt, platform, run) tuple. For a brand with P prompts running on all five engines daily, this produces $5P$ tuples per day, or $\sim 150P$ per month per brand.

4 Methodology

4.1 Prompt construction

Prompts are generated by Claude Sonnet (temperature 0.7) seeded with the brand’s category, keywords, location, reach (worldwide / country / local), and 30-day market-research context. The market-research context seeds the generator with the live questions and competitor framing current in the brand’s category, so the prompts track how the category is actually discussed rather than a static keyword list. Generation honors a category mixture ($\Sigma = 1.0$ per §3.4) and produces prompts at a controlled difficulty (1–5) and search-volume estimate (1–5). Difficulty captures how specialized a query is and search-volume how commonly it is asked, so a brand’s coverage can be read across both high-demand head queries and rarer long-tail ones. By default the generator produces *unbranded* prompts, keeping the brand name and its aliases out of the prompt text. This is what lets us measure *generic* category visibility rather than self-referential mentions, and it is the unbranded measure we lean on throughout the paper. Branded prompts are produced only when a user explicitly opts into the comparison or brand-research categories.

4.2 Visibility, position, sentiment

For a prompt q issued to platform p , the engine returns response $R(q, p)$. We measure:

$$\begin{aligned} m(q, p) &\in \{\text{fr, t3, m, i, none}\} \\ \pi(q, p) &\in \mathbb{N} \cup \{\infty\} \\ \sigma(q, p) &\in \{\text{pos, neut, neg}\} \\ S(q, p) &= \{(\text{url, domain, type, mb})\} \end{aligned}$$

where m is mention-type (first_rec, top_three, mentioned, indirect, none), π is the ordinal position, σ is sentiment, and S is the set of cited sources.

Aggregations of interest:

$$V_p(Q) = \frac{|\{q \in Q : m(q, p) \neq \text{none}\}|}{|Q|} \quad (1)$$

$$\bar{P}_p(Q) = \text{mean}\{\pi(q, p) : \pi(q, p) < \infty\} \quad (2)$$

$$\text{SoV}_p(b) = \frac{M(b, p)}{M(b, p) + \sum_{c \in \mathcal{C}(b)} M(c, p)} \quad (3)$$

where $M(b, p)$ is the count of mentions of brand b on platform p , and $\mathcal{C}(b)$ is the set of *qualified* competitors of b — those that recur or resolve to a real domain, so one-off names an engine hallucinates do not inflate the denominator (§3.4). A practitioner-level treatment of the share-of-voice measurement choices behind Eq. (3) — denominator construction, position weighting, and sample averaging — is published separately (Ranqo, 2026j).

4.3 Source overlap

For prompt q , let $D_p(q)$ be the set of cited domains derived from $S(q, p)$. The **Jaccard overlap** of two platforms p, p' on prompt q is

$$J(p, p'; q) = \frac{|D_p(q) \cap D_{p'}(q)|}{|D_p(q) \cup D_{p'}(q)|}. \quad (4)$$

Cross-platform agreement at the source level is the average Jaccard over all pairs of platforms and all prompts.

4.4 Closed-loop lift

For an ImprovementRecommendation r targeting dimension d of URL u , with $s_d(\cdot)$ denoting the dimension score:

$$\begin{aligned} b(r) &= s_d(u, t_0) \\ c(r) &= s_d(u, t_1) \\ \ell(r) &= c(r) - b(r) \end{aligned}$$

where b, c, ℓ are baseline, current, and lift respectively. We report mean $\ell(r)$ over implemented recommendations per dimension. This is a per-treatment-effect estimate without a randomized control. We discuss the resulting selection bias in §8.

4.5 Statistical conventions

Uncertainty on means and slopes is reported with a non-parametric bootstrap (10,000 resamples with replacement). For the three-tier visibility comparison (§6.1) we use a Kruskal–Wallis omnibus test across tiers, followed by pairwise Mann–Whitney U tests with Bonferroni correction and Cohen’s d as the effect size, and we check the small Tier 1 cell with a leave-one-out sensitivity. Per-engine trajectories (§6.4) are summarized by an ordinary-least-squares slope of visibility on run index, fit per (brand, engine) and averaged across brands with a bootstrap CI. We prefer nonparametric tests because the per-brand visibility distributions are bounded and right-skewed rather than normal.

5 Reference Study: CRM Category, Five Platforms, January 2026

Before the production cohort, it helps to look at one category up close. We summarize Ranqo’s published CRM study (Ranqo, 2026b): 50 unbranded prompts across the six categories, sent to all five engines ten times (2,500 responses, 9,600+ brand mentions), tracking ten CRM brands (Salesforce, HubSpot, Zoho, Pipedrive, Freshsales, Monday Sales CRM, Copper, Insightly, Capsule, and Close), in January 2026.

The category is top-heavy. Salesforce, HubSpot, and Zoho together took 51% of all CRM mentions, while the bottom three (Copper, Insightly, Capsule) showed up in fewer than one answer in five. The engines also disagree on who comes first. Average mention position when mentioned (lower is better) splits clearly by engine (Table 2).

ChatGPT leans hardest toward Salesforce and likes numbered rankings, Claude spreads its mentions the most evenly, and Perplexity over-indexes on Pipedrive, which we put down to its live search picking up recent comparison reviews. The drop-off after position three is steep on every engine. Source-link inclusion varies just as much (Perplexity 95%, Gemini 35%, Grok 20%, ChatGPT 15%, Claude 10%), and sentiment skews positive for the leaders (HubSpot 71% positive) while the bottom tier carries more neutral and negative framing.

The takeaway sets up the rest of the paper: measurement has to be done per platform, a 50-prompt slice is already enough to see category concentration, and the per-platform quirks line up with known differences between the engines.

6 Empirical findings from production data

The dataset behind this section is drawn from Ranqo’s production database: **102 brands, 3,508 completed tracking runs, 102,025 prompt responses, roughly 15,815 brand mentions, and 149,912 source citations across five engines**, gathered between March and May 2026. We report six findings, leading with the headline:

the brand-stature visibility ladder (§6.1) — the one with inferential support and no counterpart in the GEO benchmarks we surveyed — then day-one recognition (§6.2), how visibility splits by prompt category (§6.3) and drifts over runs (§6.4), where the citations come from (§6.5), and how much noisier sentiment is than mention (§6.6).

Every number below was produced by SQL run against the production schema; the analysis methods are summarized in the reproducibility appendix.

6.1 The brand-stature visibility ladder

Brand stature is the dominant determinant of day-1 AI visibility, and it is this paper’s headline result: **each step down the stature ladder costs roughly 30 percentage points of unbranded visibility** — Tier 1 brands appear in 73% of unbranded category answers on the first run, Tier 2 in 44%, and Tier 3 in just 11%. We find no comparable multi-tenant quantification of this effect in the GEO benchmarks we surveyed (Aggarwal et al., 2024; Puerto et al., 2025; Yang, 2025). We isolate *which brands the engines cite* and quantify the headroom per tier. The reading is cross-sectional and observational — stature is not randomized — so we report it as a quantification, not a causal claim.

Day-1 visibility forms a clean ladder. Mean per-brand prompt-level visibility on run 1 (each brand’s first completed tracking run), restricted to *unbranded category prompts* (where the brand name does not appear in the prompt text). Branded prompts inflate every tier by construction and are reported separately for comparison only:

Tier	n	Unbranded	95% CI	All*
Tier 1	11	72.9%	[60.1, 84.2]	73.5%
Tier 2	36	43.6%	[36.4, 50.9]	52.2%
Tier 3	55	11.4%	[4.2, 20.3]	17.3%

* “All prompts” includes a $\sim 9\%$ subset of branded queries (e.g., “what does Razorpay do”) that recognize the named brand by construction at $\geq 94\%$ across every engine. The inflation is largest in Tier 2 (+8.6 pp) because mid-cap brands have the most branded-query coverage in their prompt sets; Tier 1 brands are recognized either way (+0.6 pp inflation).

Bootstrap CIs use 10,000 resamples with replacement at the brand-cohort level on the unbranded subset. A Kruskal–Wallis test rejects equality of unbranded tier distributions ($H = 38.32$, $df = 2$, $p = 4.78 \times 10^{-9}$). Pairwise Mann–Whitney comparisons survive Bonferroni correction at $\alpha = 0.05/3$ in all three directions: Tier 1 vs Tier 2 $U = 343$, $p_{\text{Bonf}} = 8.1 \times 10^{-4}$, Cohen’s $d = 1.57$; Tier 1 vs Tier 3 $U = 575$, $p_{\text{Bonf}} = 8.3 \times 10^{-6}$, Cohen’s $d = 2.34$; Tier 2 vs Tier 3 $U = 1629$, $p_{\text{Bonf}} = 6.4 \times 10^{-7}$, Cohen’s $d = 1.31$. All three effect sizes are large by Cohen conventions ($d > 0.8$); the Tier 1-vs-Tier 3 effect is enormous. The per-tier steps are 29.3 pp (Tier 1→Tier 2) and 32.2 pp (Tier 2→Tier 3), and the difference is not statistically marginal.

Table 2. CRM study: average mention position by brand and engine (lower is better).

Brand	ChatGPT	Gemini	Perplexity	Claude	Grok
Salesforce	1.2	1.3	1.4	1.8	2.1
HubSpot	1.5	2.0	1.8	1.3	1.9
Zoho	3.1	3.8	3.5	2.4	3.2
Pipedrive	3.4	3.6	2.8	3.1	4.1

We report the unbranded-only number as the primary metric because it matches the measurement question a brand faces in GEO, namely whether the brand is cited when a user asks for recommendations in the category without naming any company. Branded coverage is a useful auxiliary signal, since it confirms the brand exists in the engine’s index at all, but it does not measure competitive recognition.

Classification rubric. We assigned each cohort brand a tier deterministically using the hand-coded rubric documented in full in Appendix B. In brief: Tier 1 requires a high-coverage Wikipedia article + recent top-tier press + public/Series-C-or-later status; Tier 2 requires Wikipedia + Series B+ funding or top-3-in-category regional leadership; Tier 3 is everything else with an established web presence; Tier 4 captures single-run and test-data records and is excluded from this section. After deduplication by domain at the May 30, 2026 snapshot the ladder cohort is $n = 11/36/55$ for Tiers 1 / 2 / 3. Appendix B lists representative Tier 1 and Tier 2 brands; a formal inter-rater audit with Cohen’s κ is left to v1.1, and the subjectivity caveat is flagged in §8.

Robustness to re-classification. The Tier 1 cohort is small ($n = 11$), so we ran a leave-one-out sensitivity on the unbranded headline. Dropping each Tier 1 brand in turn shifts the Tier 1 unbranded mean to between 70.5% and 76.7%, a range of -2.4 to $+3.8$ percentage points around the headline 72.9%. The headline is therefore not driven by any single Tier 1 brand and is conservative against borderline reclassification. The three-tier ordering also holds when the single-run prospect brands (Tier 4) are folded in: the step structure is preserved and the Tier 3 estimate barely moves (to about 10%) on $n = 191$ rather than 55.

Small brands are a floor problem on every engine. Across small brands — those outside the Tier 1 and Tier 2 roster, including the single-run prospect brands (Tier 4 of Appendix B) — day-1 unbranded recognition is low and similar across the major engines: ChatGPT 9.9%, Gemini 8.9%, and Perplexity 6.8% ($n = 169$ – 187 per engine). Small brands are hard to surface everywhere, not specifically on retrieval-grounded engines.

6.2 Day-1 recognition: brands surface immediately when named, far less when not

Practitioner folklore says it takes around six months for a brand to become visible to AI engines. Our data does not support that timeline, with one qualification: the answer depends on whether the prompt names the brand.

Day-1 first-run recognition rate (per-prompt). For each brand’s first completed tracking run, we compute the share of prompts on which the brand was mentioned, per engine, split by whether the prompt is branded (the brand name appears in the prompt text) or unbranded (e.g., “best Indian fintech payment platform”).

Engine	Branded	Unbranded	All prompts*
ChatGPT	94.2%	22.1%	28.6%
Gemini	94.0%	18.7%	26.6%
Perplexity	98.5%	23.9%	30.9%
Claude	100.0%	51.5%	54.6%
Grok	100.0%	12.0%	24.1%

* “All prompts” is the per-prompt aggregate across both subsets; branded prompts are $\sim 9\%$ of first-run responses but recognize at $\geq 94\%$ on every engine, which pulls the aggregate up.

The branded column is largely trivial: when a prompt asks “what does *Razorpay* do,” *Razorpay* is mentioned by construction. The **unbranded column is the GEO-relevant measure**: it asks whether the engine surfaces the brand in answer to a category-level question where the user does not name it. Two things stand out. First, **Claude posts the highest unbranded recognition** — over half of category-level prompts return a mention of the tracked brand on the first run — though with a sampling caveat: Claude and Grok are Agency-tier engines, so they run on far fewer brands than ChatGPT, Perplexity, and Gemini, and that smaller cohort skews toward higher-stature brands. Claude’s high rate therefore partly reflects which brands it sees (higher-stature brands surface more; §6.1) rather than the engine alone. Second, the other four engines sit in the 12–24% regime, far below the branded ceiling and consistent with the brand-stature ladder in §6.1: most unbranded recognition is concentrated in higher-stature brands.

One caveat concerns brand-new entities. A brand with no prior web presence can take longer to surface, but that is a different regime from the established brands a Ranqo customer typically tracks.

6.3 Visibility by prompt category

Average per-brand visibility differs sharply by prompt category (Table 3). We compute it across all tracked brands (a broader set than the 102-brand cohort used for the trajectory and ladder analyses, since a category average needs only a single run to contribute), as the mean across brands of each brand’s mention rate within a category.

Two things stand out among the unbranded categories. Broad *discovery* questions (“best running shoes”) surface a brand most often, at 23%, while specific *problem* and *use-case* questions are the hardest, at roughly 11%. A brand can be findable for the broad ask and nearly invisible for the specific one. Claude leads clearly on unbranded prompts (34–47% versus 9–22% on ChatGPT, Perplexity, and Gemini); Grok looks similar but its per-category sample ($n = 11$ –15) is too small to weigh.

6.4 Persistence: mostly flat, with a mild decline on ChatGPT and Perplexity

Per-engine visibility slope over runs. For every brand with at least two weeks of tracking on a given engine, we fit a slope of per-prompt visibility on *unbranded* category prompts against run-index per (brand, engine), then report the mean across brands with a bootstrap 95% CI (10,000 resamples):

Engine	Mean slope	95% CI	n
ChatGPT	−1.34%/run	[−2.45, −0.38]	101
Gemini	−1.09%/run	[−2.80, +0.23]	95
Perplexity	−0.76%/run	[−1.66, −0.03]	98
Claude	−0.52%/run	[−1.37, +0.22]	25
Grok	+0.03%/run	[−0.41, +0.34]	11

On unbranded prompts the picture is mostly flat, but not entirely. ChatGPT (−1.34%/run, CI [−2.45, −0.38]) and Perplexity (−0.76%/run, CI [−1.66, −0.03]) both show a real downward drift: their CIs exclude zero. Claude, Gemini, and Grok stay statistically flat (their CIs cross zero). The measured band runs from −1.34% to +0.03% per run. So a brand that does nothing tends to lose unbranded visibility slowly on ChatGPT and Perplexity while holding steady on the rest. This is a slow erosion, not the dramatic “AI forgets you” collapse, and some of it may be prompt-set drift (§8) rather than true decay; either way it argues for active upkeep over set-and-forget.

Reading the slopes as a no-intervention baseline.

We do not read the downward drift as evidence that visibility cannot be moved. The more plausible explanation is that very few brands in our cohort actively tried to move it during the observation window. The cohort consists of brands that signed up for tracking and let the platform observe them; in almost all cases, no explicit content, authorship, or page-structure intervention was applied based on Ranqo’s recommendations. We therefore read these trajectories as the *no-intervention baseline*: left alone,

unbranded visibility holds on most engines and slips slowly on ChatGPT and Perplexity. Whether a deliberate intervention can bend these trajectories upward is the v1.1 RCT question (Protocol P3, §6.7), not something this observational baseline can settle.

Do the lower tiers climb on their own? Mostly not, at least not visibly inside our window. We fit the same per-run slope within each (tier, engine) group: almost every group’s 95% confidence interval crosses zero, echoing the flat cohort slopes above. A few individual brands did move a lot on their own, each climbing sharply on a single engine, but those are single-brand series with no confidence intervals, picked out after the fact, so we treat them as anecdotes rather than evidence.

Mention behavior is deterministic but lopsided toward invisibility. For every (brand, prompt, platform) cell tracked across at least 3 runs on unbranded category prompts, we classify the mention pattern:

Pattern	Share
Never mentioned	63.2%
Always mentioned	14.3%
Mostly mentioned ($\geq 70\%$)	7.0%
Rarely mentioned ($< 30\%$)	8.7%
Flipping (30–70%)	6.8%

On unbranded prompts, AI surfacing is near-binary: 77.5% of cells are strictly always- or never-mentioned and 93.2% sit outside the volatile 30–70% band, with only 6.8% flipping. A cell is usually never mentioned (63.2%) or always mentioned (14.3%), closer to a fixed property of the (brand, prompt, engine) cell than a coin-flip. The dominant pole is invisibility: for most (brand, prompt, platform) cells the brand simply fails to surface, run after run, rather than appearing intermittently. This skew tracks the cohort’s stature mix — 11, 36, and 55 brands across Tiers 1, 2, and 3 (§6.1) — since the Tier-3 majority rarely surfaces on unbranded category prompts and so lands at the never-mentioned pole. Because so few cells flip, a single tracking run places a cell on the right side of the stable/volatile split the large majority of the time; the remaining 22.5% (mostly-mentioned, rarely-mentioned, and flipping combined) are where measurement noise concentrates, and the conservative reading is that mention-rate estimates need ≥ 3 runs to settle for cells in the middle.

ParallelDots: divergent trajectories by engine. To make the per-engine divergence concrete, we walk through a single brand, ParallelDots (a mid-cap series in our dataset), as an illustrative case rather than evidence: one brand cannot support a population-level claim, but it shows in a single record how the same brand and the same prompts can pull apart across engines — which is what motivates tracking each engine separately. Per-week

Table 3. Average per-brand visibility by prompt category (macro-average across brands), overall and per engine. Comparison and brand-research prompts name the brand in 46% and 91% of cases respectively, which inflates their mention rate; the four categories above the rule are under 1% branded.

Category	Overall	ChatGPT	Gemini	Perplexity	Claude	Grok
Discovery	22.9%	22.0%	20.6%	20.0%	47.2%	25.1%
Problem/solution	10.8%	10.4%	8.9%	9.1%	36.9%	22.1%
Use case	11.4%	9.9%	11.9%	10.3%	33.7%	28.0%
Expert	15.3%	15.5%	12.8%	12.6%	40.0%	38.3%
Comparison	74.6%	73.3%	72.6%	73.7%	76.4%	53.4%
Brand research	97.0%	97.9%	95.7%	95.7%	98.8%	94.4%

Per-category n is 156–241 brands (59–73 for the two branded categories). Claude and Grok are Agency-tier engines, so their per-category n (Claude 26–35, Grok 11–15) is far smaller than ChatGPT, Perplexity, and Gemini (150–240) and skews toward higher-stature brands.

mention rate (all prompts) over its first five complete tracking weeks:

Engine	wk 0	wk 1	wk 2	wk 3	wk 4
ChatGPT	45%	43%	34%	27%	20%
Perplexity	37%	38%	43%	48%	62%
Claude	24%	24%	23%	26%	26%

The engines pull apart over the five weeks: Perplexity climbs from 37% to 62%, ChatGPT slides from 45% to 20%, and Claude stays flat in the mid-20s — same brand, same prompts, the two engines that move going in opposite directions. This is exactly the per-engine divergence the v1.1 RCT design (Protocol P3, §6.7) is built to test at population scale, where the cohort slopes above — not a single record — can settle it.

6.5 Source composition: corporate dominance is mostly third-party brands, not the brand’s own pages

Across 149,912 citations drawn from mention-bearing prompts, the source-class distribution is:

Source class	Citations	Share
Corporate / third-party brand pages	112,763	75.2%
Youtube / video	6,311	4.2%
Tech & business media	5,666	3.8%
Reddit / community forums	4,992	3.3%
Academic + Govt. + Social + Developer	4,617	3.1%
Brand-owned (own domain)	4,392	2.9%
Wikipedia / reference	3,882	2.6%
Software review (G2 / Capterra)	1,604	1.1%
Other (long tails)	5,685	3.8%

The split is the finding: only 2.9% of citations point at the brand’s own domain, while 75.2% point at *other* companies in the same space. We label this dominant class *corporate / third-party brand pages*: commercial websites owned by businesses other than the tracked brand — competitors, peers, and vendors in the same category — including their homepages, product and landing pages, and corporate blogs. It is distinct from *brand-owned*, which is the tracked brand’s own domain. Own and third-party corporate pages together are about 78% of all 149,912 citations. AI engines preferentially construct “alternatives”

answers, and in those answers the dominant source is a set of peer-brand product pages, not the user’s own site. The Matthew-effect dynamic that Algaba et al. (2025) documented for scientific citation replicates at the brand level: already-cited domains pull more citation. This peer-brand bucket is the largest competitive-landscape signal we measure.

Among non-corporate sources, video leads. Once corporate pages are set aside, the largest non-corporate source is video: YouTube is cited in 4.2% of citations, ahead of editorial tech and business media (3.8%), Reddit and other community forums (3.3%), and Wikipedia (2.6%). This reorders a common assumption. Community forums — often reported as the leading non-corporate AI-citation channel, particularly on Perplexity — here sit behind both video and editorial media, though they remain a meaningful surface a brand cannot ignore. Wikipedia, often treated as the reference anchor, is cited less often than video, media, or forums. A brand absent from video and editorial coverage is missing the two largest non-corporate citation surfaces these engines draw on.

The long tail. The remaining 3.8% is a long tail of many small, individually-rare domains that match no named class: niche or regional company sites, small and personal blogs, and miscellaneous pages on uncommon top-level domains. No single domain in this bucket is frequent. We keep it explicit as *other (long tails)* rather than folding it into the corporate class, so the corporate share is not overstated.

Page types: most cited pages are articles, and the listicle leads them. Beyond *which site* an engine cites, we also label each cited URL by *what kind of page* it is. Every cited page is either *content* — a page that informs, such as an article, guide, or video — or a *non-content* page such as a homepage, product page, or other landing page. The two cuts are independent: a competitor’s blog post, for example, counts as a corporate site in the source cut but as a content page here, and the content an engine cites can equally come from a media outlet, a review site, or an academic or reference page.

When an engine cites a page, about 59% of the time it is content (usually an article); the other 41% are homepages, product pages, or other non-content pages. Within that content, one format dominates: the *listicle* — a ranked “best-X” roundup such as *best CRM tools* or *top running shoes* — is 35.7% of content citations. It is followed by the *generic article* (31.0%), an article with no more specific format (a news item, an opinion piece, or a plain explainer), and the *how-to guide* (9.7%), a step-by-step walkthrough; together those three are about three-quarters of all content.

The listicle is also the highest-leverage page a brand can target: once a ranked list includes the brand, that single page becomes a source the engines reuse across many different prompts, and such lists appear on media sites, blogs, and review sites alike. The full distribution across content formats is as follows:

Content format	% of content	% of all
Listicle (ranked list)	35.7%	21.0%
Generic article	31.0%	18.3%
How-to guide	9.7%	5.7%
Generic video	4.6%	2.7%
Comparison	4.5%	2.6%
Topic guide	3.9%	2.3%
Documentation	3.3%	2.0%
Review	2.6%	1.5%
Research	2.5%	1.5%
Checklist	1.5%	0.9%
Alternatives / case study	0.7%	0.4%

Content citations concentrate in a handful of formats: listicles and generic articles together are two-thirds, and with how-to guides about three-quarters of all content. The listicle is the single largest format, and its 35.7% is a lower bound — it counts ranked lists in any category, not just software.

6.6 Measurement reliability: sentiment is $6.7\times$ noisier than mention

A scoring system that combines mention rate with sentiment must account for the fact that the two signals have different noise floors. For (brand, prompt, platform) cells with at least 3 mentions on unbranded prompts, the sentiment pattern across runs distributes as:

Pattern	Share
Always positive	29.9%
Positive/neutral mix	21.8%
Always neutral	2.3%
Negative/neutral mix	0.5%
Always negative	0.0%
Flipping (pos $\leftrightarrow$ neg)	45.5%

The flipping rate is 45.5% for sentiment and 6.8% for mention (§6.4). Sentiment is therefore $6.7\times$ noisier than mention at the same observational level. Operationally, sentiment-weighted scores need a much larger denominator (we use ≥ 10 prompts per platform per brand) before they stabilize.

The same data carries a second, sharper observation: not a single cell is consistently negative (0.0%). AI engines

effectively never settle into a negative characterization of a tracked brand — negativity, when it surfaces at all, is transient rather than systematic.

6.7 Designed but not yet reported: the v1.1 protocol slate

We close the empirical section with seven protocols. Each is designed against the same production data but needs more data, controlled variation, or analysis we have not finished. We publish them so others can run or contest them, and to lay out a clear agenda for follow-up work.

P1. Cross-platform source overlap. Mean pairwise Jaccard overlap of cited sources across the five engines on the same prompt. On the CRM study it is about 0.12; v1.1 widens it to 1,000 prompts and tests it against the cross-platform-disagreement results of Yang (2025); Kirsten et al. (2025).

P2. Position-decay distribution. How mention position is distributed when a brand is cited at all, per platform, over a full quarter. We expect a top-3 share of 60–75% in concentrated categories and 40–55% in long-tailed ones.

P3. Closed-loop lift after recommendation (RCT). Pair comparable brands within tier and category, randomly assign one of two recommendations, and re-audit at $t_1 \geq 14$ days. Aggarwal et al. (2024) saw 25–40% lifts from content-quality changes; we expect dimension-level lifts strongest on Authority & Trust, AI Readiness, and Citation Potential. This is the centerpiece of the planned follow-up paper, which will report the visibility lift a customer brand can actually expect.

P4. Schema vs. citation. Regress brand visibility on structured-data, AI-readiness, and content-quality scores per engine, expecting a near-zero independent schema effect. Ranqo already treats schema as hygiene rather than a citation lever (Ranqo, 2026i); P4 tests that.

P5. Entity-first sequencing. Test whether pages that state their entity and provenance early get cited more than pages with high overall content quality but weaker entity-first structure, building on the positional findings in Aggarwal et al. (2024).

P6. Web-search on/off. Claude and ChatGPT run web search weekly per brand. Comparing search-on against search-off visibility within the same brand separates retrieval-layer from training-layer visibility at scale. We have the data; the analysis is pending.

P7. White-hat C-SEO replication. Puerto et al. (2025) found prompt-engineering tactics mostly fail. We replicate this with audit-dimension improvements instead: one arm targets AI Readiness, Citation Potential, and Authority & Trust, the other targets Crawlability or Page Speed, compared on visibility change.

6.8 Summary

Table 4 collects the six reported findings and the seven designed-but-unrun protocols. The headline and statistically strongest result is the brand-stature visibility ladder (§6.1): on each brand’s first tracking run, unbranded category visibility is 73% for Tier 1 brands, 44% for Tier 2, and 11% for Tier 3 — about 30 percentage points per step down. Because stature is observed rather than randomized, we report it as a multi-tenant *quantification* of an expected effect rather than a novel discovery.

Three further results are descriptive contributions the existing literature does not supply. The source-class decomposition (§6.5) finds the brand’s own domain is only 2.9% of citations, video is the largest non-corporate source, and the listicle is the single most-cited content format. The mention-determinism / runs-to-stability result (§6.4) finds AI surfacing is near-binary — 77.5% of (brand, prompt, engine) cells are strictly always- or never-mentioned — and that, left alone, unbranded visibility holds on most engines and drifts down only slowly on ChatGPT and Perplexity. And the $6.7\times$ sentiment-vs-mention noise gap (§6.6) shows the two signals cannot share one noise budget. Two supporting findings complete the set: day-1 recognition (§6.2), where a brand surfaces immediately when a prompt names it but far less when it does not, reported with explicit limits on what it supports; and the prompt-category split (§6.3), where broad discovery questions surface a brand most ($\sim 23\%$) and specific problem and use-case questions least ($\sim 11\%$). The protocols (P1–P7, §6.7) are a research agenda, not results, and are not claimed as contributions of this paper.

7 Discussion

7.1 Four practical conclusions, especially for smaller brands

1. **Know which tier you are in before you spend on GEO.** The right GEO program looks different in each tier; we frame these as hypotheses consistent with the data, not established commercial claims. Tier 1 global brands sit at 73% day-1 visibility, effectively at saturation (the $n = 11$ Tier-1 cell makes strong sub-claims fragile); their need is share-of-voice against competitors rather than basic recognition. Tier 2 brands sit at 44% with the largest absolute headroom, but the no-intervention baseline (§6.4) shows unbranded visibility drifting flat-to-down rather than climbing on its own; whether targeted intervention closes that headroom is the v1.1 RCT question (Protocol P3), not a baseline finding. Tier 3 brands sit at 11%; the fastest Tier-3 movers we observed (§6.1) still travel only 10–20 absolute points across the window, so the highest-confidence move is brand-mass investment (Wikipedia, mainstream press, sustained YouTube presence) to clear the Tier 2 threshold before running heavy per-engine optimization. The specifics differ by sector; see our SaaS playbook (Ranqo,

2026f) and DTC guide (Ranqo, 2026g).

2. **Treat each engine as a separate market.** Cross-platform source overlap is low. Yang et al. (Yang, 2025) found this across 366,000 news citations; we find the same pattern across 102,025 brand-tracking responses. A Perplexity win does not transfer to Gemini, and the working playbook for each engine differs accordingly (Ranqo, 2026e).
3. **Put your facts up front, not your schema markup.** Aggarwal et al. (2024) showed that content-quality additions (quotations, statistics, citations) drive the largest GEO lifts. Heavy structured-data markup, by contrast, does little on its own, which is what Protocol P4 sets out to confirm. The practical move is to state your key facts clearly and early on the page, where the engine is most likely to read them; the page-level factors that most affect whether a brand gets cited are detailed in (Ranqo, 2026d).
4. **Close the loop in weeks, not quarters.** Our day-1 recognition finding (§6.2) shows established brands are recognized within days, while brand-new entities take longer. Either way the loop runs on weeks: make a change, then re-audit and iterate at 14–30 days.

7.2 The honest causal status of this dataset

The empirical section reports baselines, trajectories, and source and sentiment composition. It does not report a randomized controlled trial of the recommendation engine. We separate three claim tiers carefully.

What the data supports. The per-tier visibility ladder, the day-1 recognition, the deterministic-cell finding, and the source-composition and sentiment shares. These are direct readings of production telemetry.

What the data hypothesizes. That the Tier 3 brand-mass ceiling can be lifted with targeted content investment. This is a hypothesis derived from the data; it needs RCT-grade validation.

What the data does not yet support: any causal claim that Ranqo’s recommendations *cause* a measured visibility lift on a brand. The recommendation engine’s closed-loop machinery is built into the data model (§3.6), but population-scale closed-loop measurement is left to v1.1 (Protocol P3).

What it most clearly establishes. The sharp three-tier day-1 visibility ladder of §6.1 (robust to the Tier-1 leave-one-out reclassification). Whether intervention accelerates those trajectories is the v1.1 RCT question (Protocol P3); we do not pre-commit to a multiplier we have not derived from a controlled experiment.

7.3 A note on white-hat measurement

One stance is worth stating plainly. Ranqo measures and recommends on page-quality signals only. It does not generate hidden text, ranking payloads, or anything designed to trick an engine, and the recommendation

Table 4. Empirical findings reported in this paper (top) and designed v1.1 protocols (bottom).

Ref	Finding or protocol	Headline	Status
6.1	Brand-stature visibility ladder	Tier 1 73% / Tier 2 44% / Tier 3 11%; $\sim$ 30-pt step per tier; trajectories null at our n	Reported
6.2	Day-1 recognition	Branded $\sim$ 94–100%, unbranded 12–52% on the first run; Claude highest on unbranded (small Agency-tier sample)	Reported
6.3	Visibility by prompt category	Discovery highest among unbranded ($\sim$ 23%), problem/use-case lowest ($\sim$ 11%); Claude highest (small Agency-tier sample)	Reported
6.4	Persistence and determinism	Unbranded visibility mildly declines on ChatGPT and Perplexity, flat elsewhere; 77.5% strictly deterministic, 93.2% outside the 30–70% band	Reported
6.5	Source composition & page types	78% corporate (own 2.9%); YouTube top non-corporate; editorial content 59% of page-level cites, listicle leads it (36% of content)	Reported
6.6	Sentiment vs mention noise	Sentiment $6.7\times$ noisier than mention	Reported
P1	Cross-platform source overlap (Jaccard)	Mean pairwise $\approx$ 0.12 on CRM; widen to $N = 1,000$	v1.1
P2	Position-decay distribution	Top-3 share by category, full quarter	v1.1
P3	Closed-loop RCT after recommendation	Per-dimension lifts on implemented recs	v1.1
P4	Schema shallow-pass budget regression	β_1 near zero per engine	v1.1
P5	Entity-first declarative sequencing	Pearson r of entity-first vs citation rate	v1.1
P6	Web-search on/off natural experiment	Training vs retrieval separation at scale	v1.1
P7	White-hat C-SEO Bench replication	Arm A (E-E-A-T/AI Ready/Cite Pot) vs Arm B (Crawl/Speed)	v1.1

engine is instructed to refuse such requests. The payoff from gaming an engine is short-lived and tends to reverse on the next model update, whereas genuine content-quality work survives those updates. That is the only kind of optimization this paper endorses.

8 Limitations

We name the constraints on the empirical claims directly. A reader should weigh the headline findings against this list.

- 1. No causal claim on the recommendation engine.** Every trajectory in §6.1 is an observational baseline, not a treatment effect; closed-loop measurement on customer brands remains future work (Protocol P3).
- 2. Convenience-sampled cohort.** The brands skew toward SaaS, retail-execution, fintech, and Indian DTC, so we do not claim category representativeness. Several cuts run on small cells (the Tier 1 group is $n = 11$, and the Agency-tier engines Grok and Claude run on far fewer brands than the rest). We report n next to every aggregate and flag small-sample readings.
- 3. Manual tier classification.** The tier assignment is hand-coded from public signals of brand stature (Wikipedia, press, funding). Because those signals are themselves proxies for web prominence, the ladder should be read as *quantifying the size* of an expected effect, not as independent proof that prominence drives citation. The rubric and rosters are in Appendix B; a formal inter-rater check is left to v1.1. The day-1 ladder is robust to the reclassifications we tried (Tier-1 leave-one-out range [70.5, 76.7]%).
- 4. Platform opacity.** Every result is inferred from API

outputs. We have no direct view into how the engines ingest, retain, or retrieve.

- 5. One model variant per engine.** We measure each engine’s search-enabled production variant and hold it fixed; whether the brand-stature ladder reproduces on other model configurations is untested here.
- 6. Sentiment is a heuristic.** Sentiment is classified by a model on the spans that mention the brand, so the 45.5% sentiment-flip rate (§6.6) mixes genuine output variance with classifier noise; we have not separated the two.

9 Conclusion and Future Work

In our 102-brand cohort, brand stature is the dominant determinant of day-1 AI visibility. The five engines diverge on sources, position, and sentiment, and they respond differently to schema markup, bylines, freshness, and web-search grounding. On the trajectory side, unbranded visibility mostly holds when a brand does nothing, drifting down only slowly on ChatGPT and Perplexity over our window. In practice, this means a brand’s GEO program has to measure each engine separately, intervene on the upstream signals the engines actually retain, and re-measure on a cycle of weeks; a quarterly cadence is too slow for a medium that re-trains on its own outputs.

Three lines of future work follow:

- 1. A multi-vendor replication track.** We publish our protocols as an invitation.
- 2. An RCT-grade closed-loop study** (Protocol P3, §6.7), in the shape of Aggarwal et al. (2024), with random assignment to treatment vs. control recommendations on Ranqo’s brand fleet. Its results, the measured visibility

lift a brand can expect from acting on the platform’s recommendations, would be the focus of follow-up work.

3. **A training-vs-retrieval separation benchmark.** The web-search-on/off design (Protocol P6, §6.7) generalizes to a public benchmark.

The practitioner literature in this space runs well behind the data, and the academic literature is fragmenting faster than any single team can synthesize it. We think the most useful thing GEO research can do in 2026 is publish protocols that others can run and numbers that others can re-score, so that claims get falsified rather than repeated.

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A Reproducibility

The statistics in §6 were computed from Ranqo’s production database with a data-dump and analysis pipeline that applies the methods of §4: nonparametric bootstrap confidence intervals, a Kruskal–Wallis omnibus test, pairwise Mann–Whitney U tests with Bonferroni correction, Cohen’s d , ordinary-least-squares trajectory slopes, and a leave-one-out sensitivity check, together with per-experiment SQL queries and the tier-classification rubric of Appendix B. Because the analysis runs against commercial customer data, the database and analysis code are not publicly released.

B Brand-tier classification rubric and roster

Rubric. The four-tier classification used in §6.1 is hand-coded from external public knowledge of brand stature, applied by the author against the underlying public-data sources, and is deterministic given the same public-data inputs.

- **Tier 1 (Global household names)** — A brand qualifies if all three of: (a) English Wikipedia article of $\geq 5,000$ words, (b) mainstream-press coverage in the last 12 months in at least three of the top-20 US/UK English outlets, (c) public-company status, $\geq$ Series C funding, or $\geq$ \$100M in annual revenue (ARR for subscription businesses).

- **Tier 2 (Established mid-cap / regional leader)** — A brand qualifies if either (a) Wikipedia article + Series B+ funding (or equivalent institutional credibility), or (b) top-3 player in a category Ranqo tracks in a specific country/region.
- **Tier 3 (Small / niche)** — Brands not meeting Tier 1 or Tier 2 criteria but with an established web presence (≥ 1 owned domain).
- **Tier 4 (Single-run / test data)** — ≤ 1 completed tracking run or evident test-data category strings, predominantly admin-created prospect brands added for a one-off evaluation. Excluded from the trajectory and ladder analyses, but included in the broader cross-brand set used for the prompt-category distribution (§6.3).

Tier 1 roster ($n = 11$, deduped by domain at the May 30, 2026 snapshot; ≥ 1 completed run). Representative brands include Stripe, Nike, Samsung, Notion, and Mailchimp, among others.

Tier 2 roster ($n = 36$, deduped). Representative brands include Olipop, Razorpay, ParallelDots, Klaviyo, Zepto, and Vuori, among others.

Tier 3 roster. The remaining $n = 55$ unique brands not in the above lists.

Inter-rater agreement. A formal second-rater audit with Cohen’s κ on a held-out sample is left to v1.1 (§6.7); the rosters in this appendix reflect single-rater coding by the author.